

Corentin Sautier

PhD Student in Computer Vision

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Education

- 2022–Today **PhD student**, *valeo.ai & ENPC, Imagine*, Paris.
Self-Supervised learning for sparse 3D point clouds using multimodal signals
- 2020–2021 **M2 in Applied Mathematics, Vision and Learning (Master MVA)**, *ENS Paris-Saclay*, Saclay.
Courses in Convex Optimization, Computational Statistics, 3D Computer Vision, Optimal Transport, Random Matrix Theory, Point Cloud and 3D Modeling, Kernel Methods, Discrete Learning and Deep Learning.
- 2017–2021 **Master's Degree in Science and Executive engineering**, *Mines Paristech*, Paris.
Courses in Mathematics, Mechanics, Computer Science, Probability, Statistics, Physics, Automatics and Climate issues. Specialization in Robotics, Computer Vision and applied Mathematics. Projects in Deep Learning, Robotics, and 3D Reconstruction.
- 2018 **Research Exchange Semester**, *Massachusetts Institute of Technology*, Boston.
Research in Monocular SLAM, and robust SLAM in variable environments. Attended courses and conferences in Computer Vision and Robotics.
- 2015–2017 **Preparatory Class in Mathematics and Physics**, *Lycée Louis Pasteur*, Neuilly-sur-Seine.
Major courses in Mathematics and Physics, Minor courses in Computer Science and Engineering Sciences.
- 2015 **High School Diploma with high honors**, *Lycée Rocroy Saint Vincent de Paul*, Paris.

Experience

Professional Experience

- 2021–2022 **Research scientist**, *valeo.ai*, Paris, France.
Research position before the start of a PhD, with focus on self-supervised representation learning for point clouds.
- 2021 **6-months Internship**, *valeo.ai*, Paris, France.
Research focused on self-supervision for point clouds, using contrastive methods between camera and LiDAR representations of a scene.
Detailed achievements:
 - Reimplemented and developed self-supervised methods on point clouds.
 - Used sparse voxel convolutions networks to reach competitive results in autonomous driving datasets.
- 2020 **6-months Internship**, *IBM Research*, Tokyo, Japan.
Research among a team in IBM's Tokyo Research Lab. The aim was to study combination of automatic planners with Deep Learning to apply Logic Programming to difficult or noisy problem setups.
Detailed achievements:
 - Learned state-of-the-art in automatic planning and model-based reinforcement learning.
 - Contributed to IBM's intellectual property with multiple patent applications.
- 2019–2020 **6-months Internship**, *Scio*, London, United Kingdom.
Start-up specialized in automatic data extraction from unformatted documents. Built text and layout classifiers to improve parsing robustness. Carried out the transition from Keras to Pytorch for the main code.

Miscellaneous

- 2025–Today **Assistant Professor.**
Giving courses and practicals on C++ and Introduction to Machine Learning
- 2023–Today **Reviewer Duties.**
Reviewer for international conferences and journals including CVPR, ICCV, ECCV, 3DV, WACV and TPAMI
Outstanding reviewer at ECCV 2024
- 2020 **School Project, Mines Paristech, Paris, France.**
3D Mesh reconstruction from Point Clouds, using Marching Cubes and implicit surface reconstruction
- 2019 **School Project, Mines Paristech, Paris, France.**
LIDAR point cloud classification algorithm using convolutional Neural Networks (U-Net).
- 2015–Today **Other.**
- Child Care for an autistic boy
 - Personal Teacher for High School Students
 - Management of an Organization, bringing fresh fruits and vegetables to students (20,000 € turnover)
 - Instrumental practice of the Cello in an orchestra & of the Guitar
 - Seasonal Employment & Blue-collar internship

Computer skills

- Python Pytorch, Tensorflow, Numpy, Scipy, Regex, Cython and other standard libraries
- C/C++ Standard libraries, OpenCV, Binding with Python
- Other $\LaTeX$, Git, OpenOffice & Microsoft Office, Linux

Languages

- French Native speaker
- English Fluent
- German Basic

Publications

- 2025 **Lidar Panoptic Segmentation: Buy the Semantic, Get the Instances for Free**, *Under review*, Corentin Sautier, Gilles Puy, Alexandre Boulch, Renaud Marlet, Vincent Lepetit.
- 2025 **UNIT: Unsupervised Online Instance Segmentation through Time**, *3DV 2025*, Corentin Sautier, Gilles Puy, Alexandre Boulch, Renaud Marlet, Vincent Lepetit.
- 2024 **Three Pillars improving Vision Foundation Model Distillation for Lidar**, *CVPR 2024*, Gilles Puy, Spyros Gidaris, Alexandre Boulch, Oriane Siméoni, Corentin Sautier, Patrick Pérez, Andrei Bursuc, Renaud Marlet.
- 2024 **BEVContrast: Self-Supervision in BEV Space for Automotive Lidar Point Clouds**, *3DV 2024*, Corentin Sautier, Gilles Puy, Alexandre Boulch, Renaud Marlet, Vincent Lepetit.
- 2023 **ALSO: Automotive Lidar Self-supervision by Occupancy estimation**, *CVPR 2023*, Alexandre Boulch, Corentin Sautier, Björn Michele, Gilles Puy, Renaud Marlet.
- 2022 **Image-to-Lidar Self-Supervised Distillation for Autonomous Driving Data**, *CVPR 2022*, Corentin Sautier, Gilles Puy, Spyros Gidaris, Alexandre Boulch, Andrei Bursuc, Renaud Marlet.
- 2020 **State Prediction in TextWorld with a Predicate-Logic Pointer Network Architecture**, *IBM Research Tokyo*, Corentin Sautier, Don Joven Agravante, Michiaki Tatsubori, KBRL at IJCAI-PRICAI 2020.
- 2020 **Towards Logical Model-based Reinforcement Learning: Lifted Operator Models**, *IBM Research Tokyo*, Corentin Sautier, Don Joven Agravante, Michiaki Tatsubori, KBRL at IJCAI-PRICAI 2020.